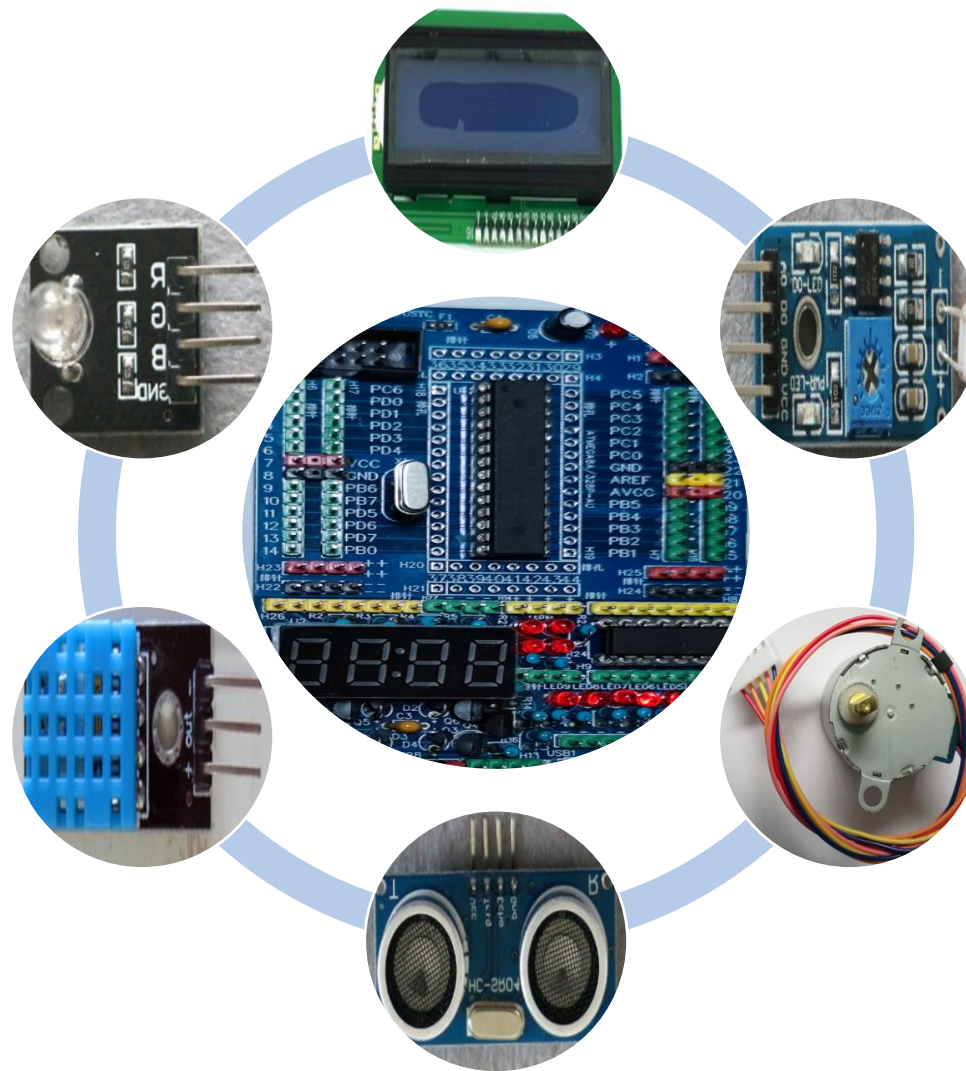


电子设计实践 基础

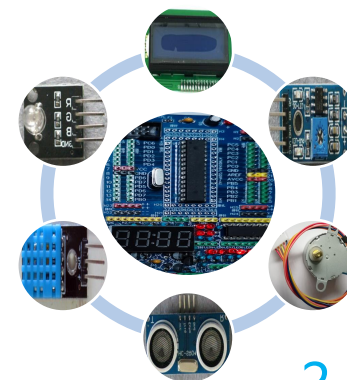
MCU与 RGB LED和触摸开关 之C语言编程



上节课内容回顾

■了解MCU芯片(ATmega8A)与其程序编写

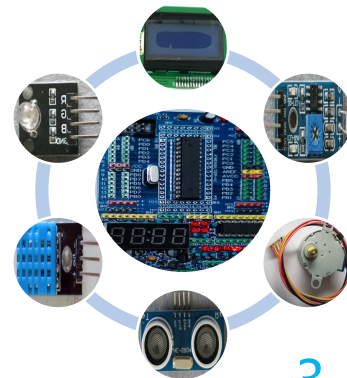
- ATmega8A的结构：8位MCU
- ATmega8A的寄存器：32, 64(IO)
- ATmega8A的指令：RISC
- ATmega8A内部存储器：8KB, 1KB, 512B
- ATmega8A的系统时钟：默认1M,2/4/8M;16M
- ATmega8A的IO端口与C语言编程
- ATmega8A的程序下载（烧写）



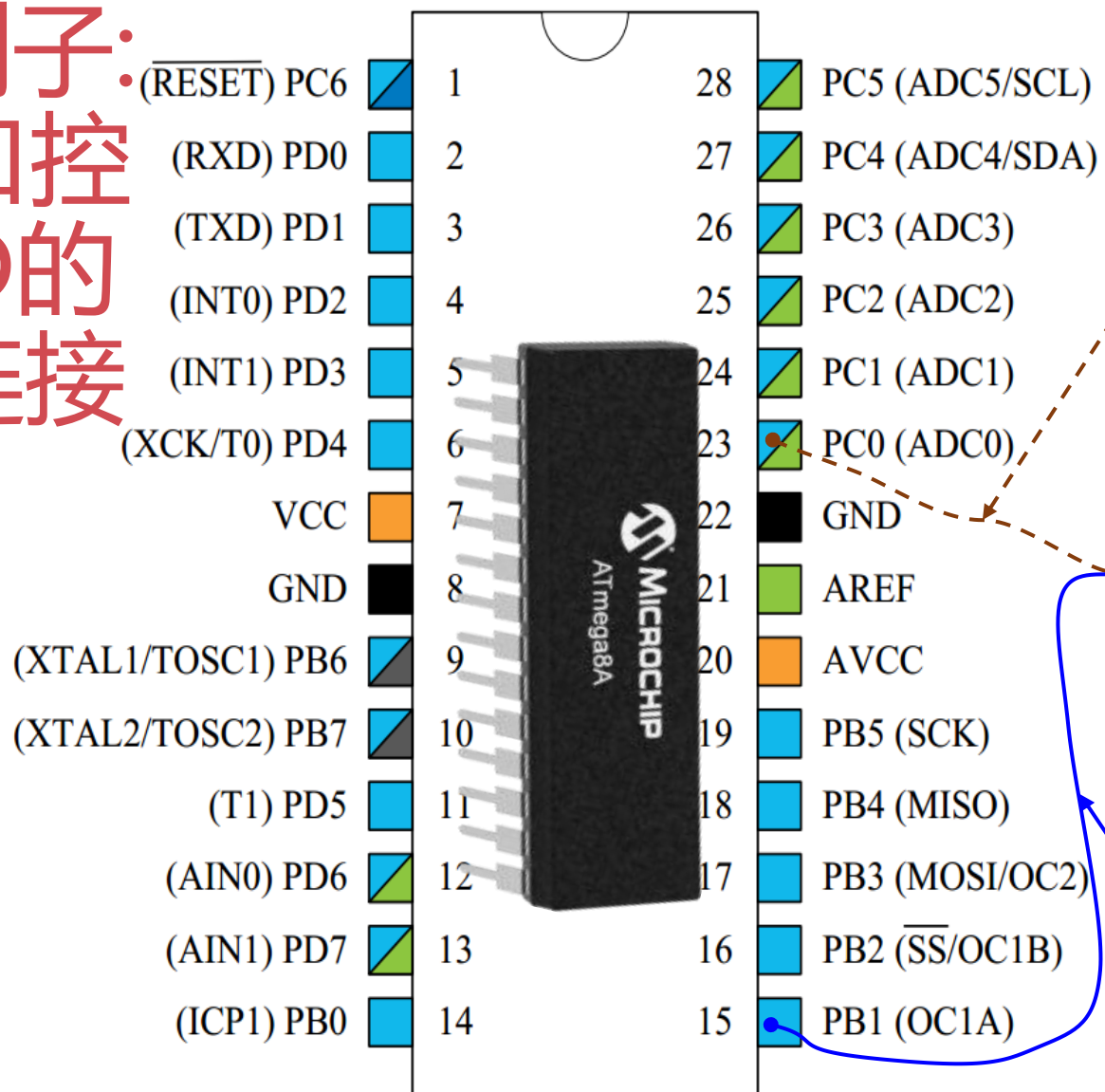
本节课主要内容

■MCU芯片与RGB LED和触摸开关及其编程

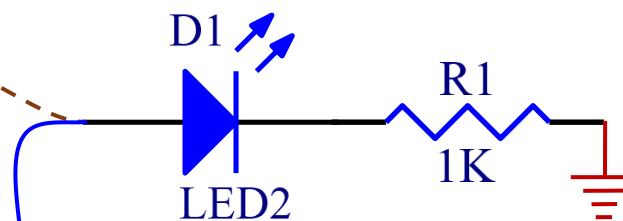
- `#include <avr/io.h>`
- IO端口控制与编程总结
- C语言的位运算
- RGB LED及其编程
- 一位触摸开关及其编程



上节例子: IO端口控制LED的 硬件连接



寄存器名	偏移地址
PORTC	0x15
DDRC	0x14
PINC	0x13



寄存器名	偏移地址
PORTB	0x18
DDRB	0x17
PINB	0x16

IO端口控制LED

```
int main(void)
```

```
{
```

```
int i,j;
```

```
*(volatile unsigned char *)(0x20+0x14) = 0x01; //DDRC[0] = 1; 0x02
```

冗长

把管脚由
PC0换成PB1

DDRB:(0x20+0x17)

```
#define DDRC *(volatile unsigned char *) (0x20+0x14)
```

```
while (1)
```

```
{
```

```
*(volatile unsigned char *)(0x20+0x15) = 0x01; //PORTC[0] = 1; 0x02
```

```
for(i=0;i<100;i++) for(j=0;j<1000;j++);
```

```
*(volatile unsigned char *)(0x20+0x15) = 0x00; //PORTC[0] = 0;
```

```
for(i=0;i<100;i++) for(j=0;j<1000;j++);
```

```
}
```

```
}
```

PORTB:(0x20+0x18)



#include <avr/io.h>

```
#include <avr/io.h>
```

```
int main(void)
```

```
{
```

```
#define DDRC *(volatile unsigned char *)(0x20+0x14)
.....
```

```
int i,j;
```

```
DDRC = 0x01; //DDRC[0] = 1
```

```
while (1)
```

```
{
```

```
PORTC = 0x01; //PORTC[0] = 1;
```

```
for(i=0;i<100;i++) for(j=0;j<1000;j++);
```

```
PORTC = 0x00; //PORTC[0] = 0;
```

```
for(i=0;i<100;i++) for(j=0;j<1000;j++);
```

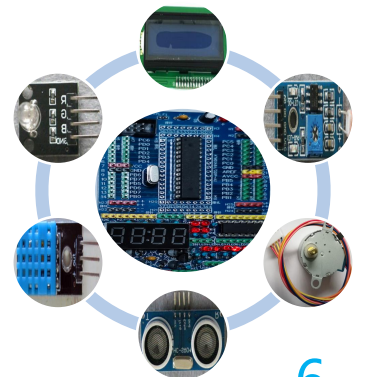
```
}
```

```
}
```

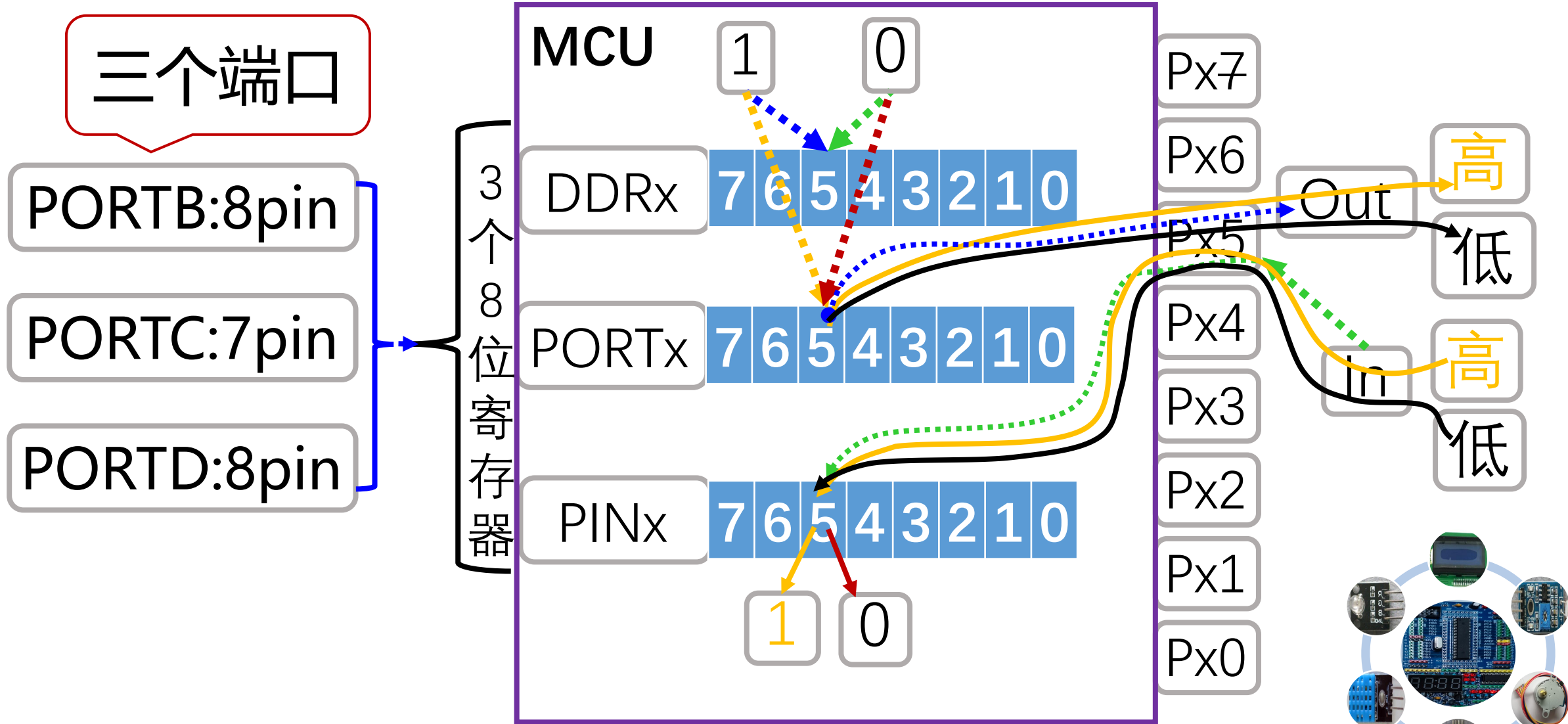
DDRB...

如管脚由PC0换成PB0?

PORTB...

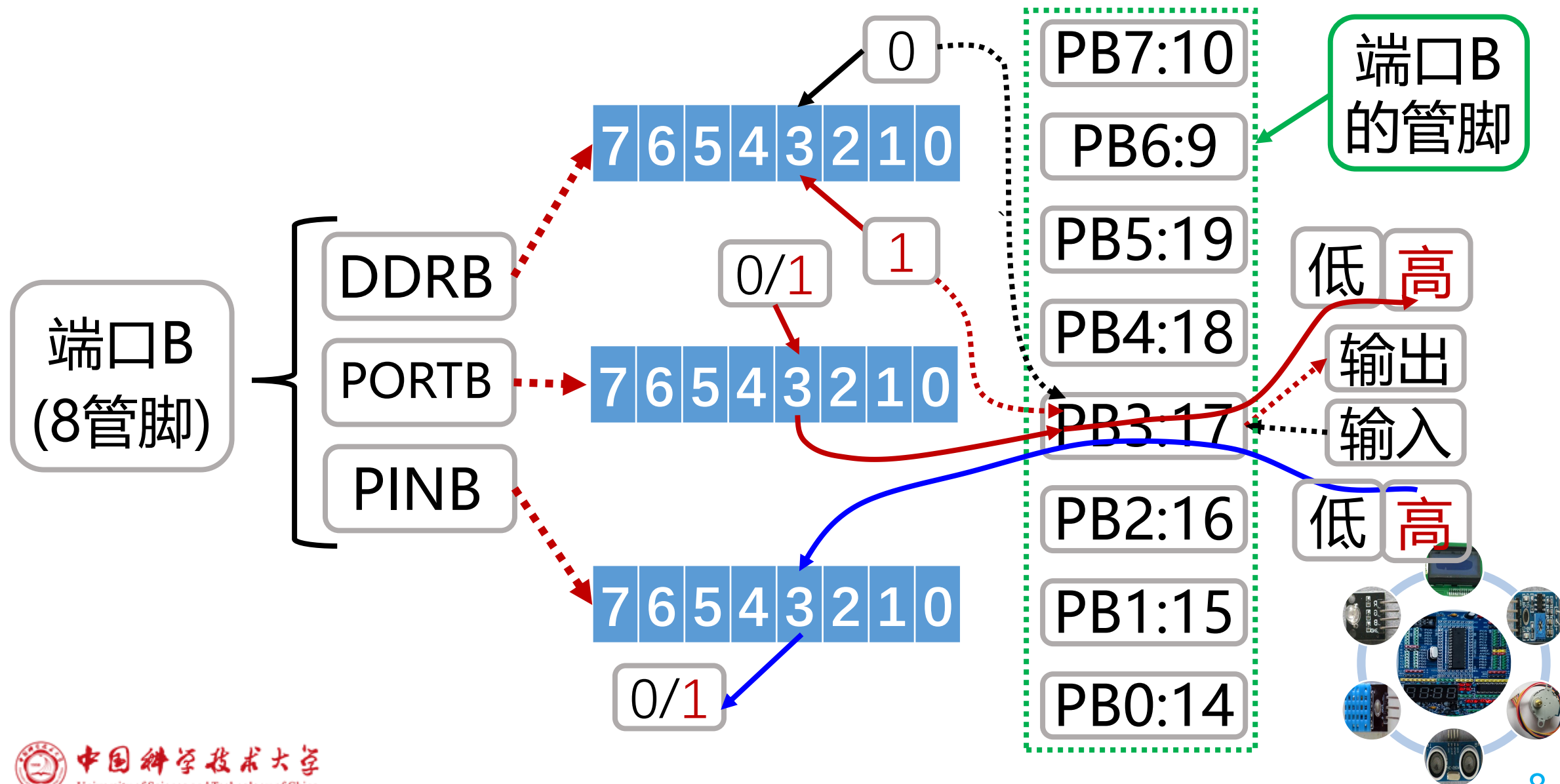


IO端口控制与编程总结

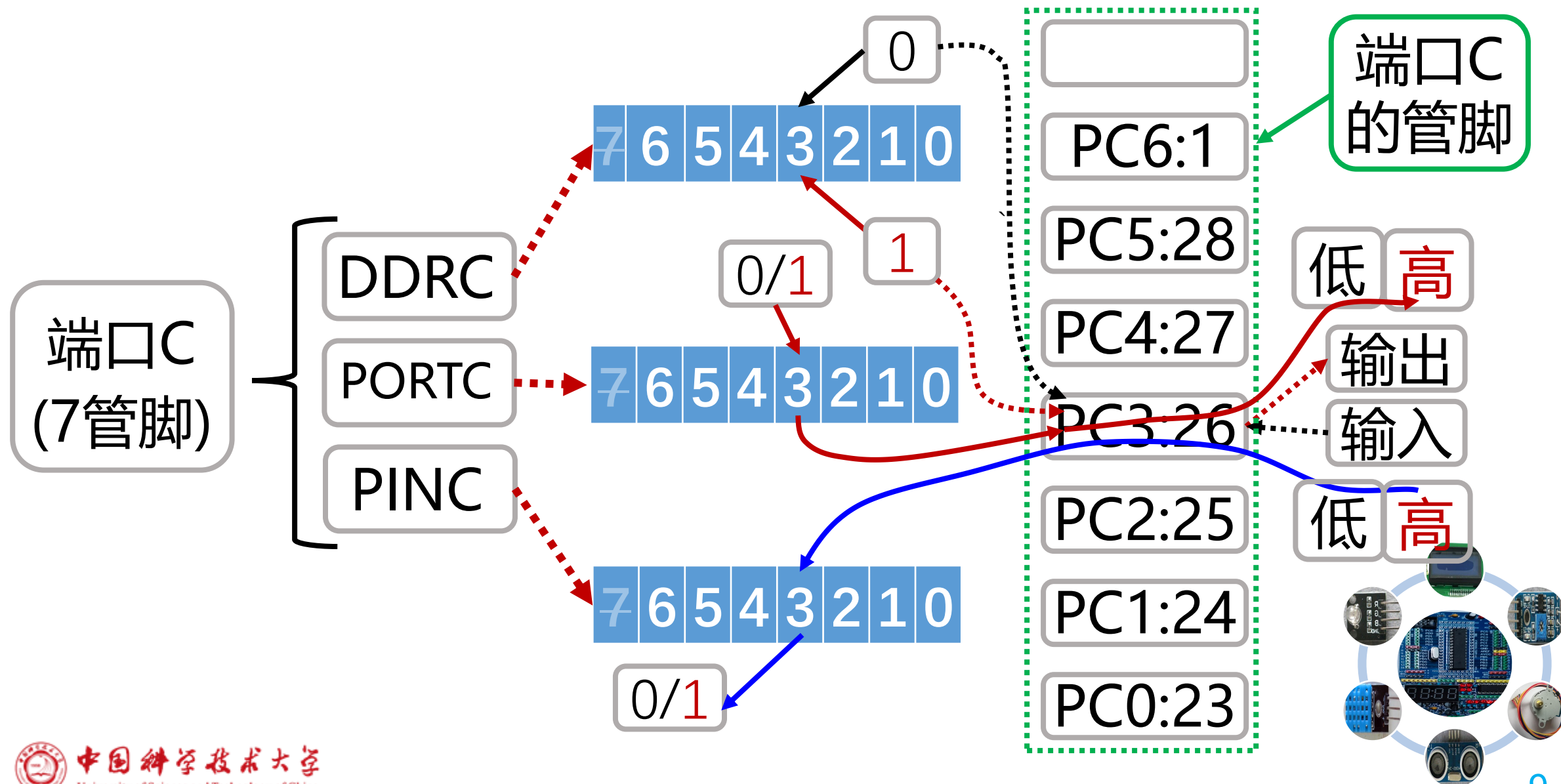


注：x为B、C或D

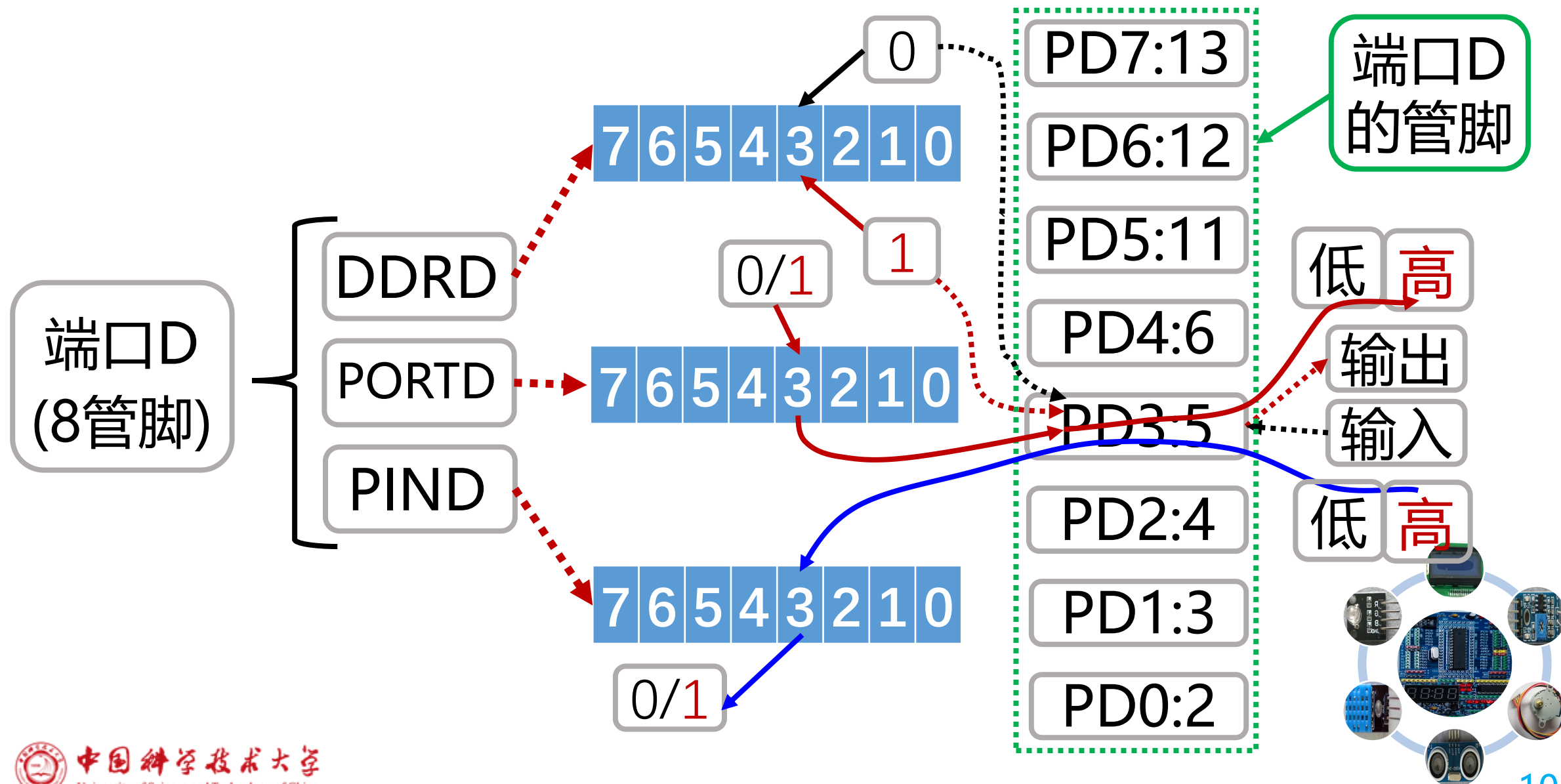
IO端口控制与编程总结：端口B



IO端口控制与编程总结：端口C



IO端口控制与编程总结：端口D



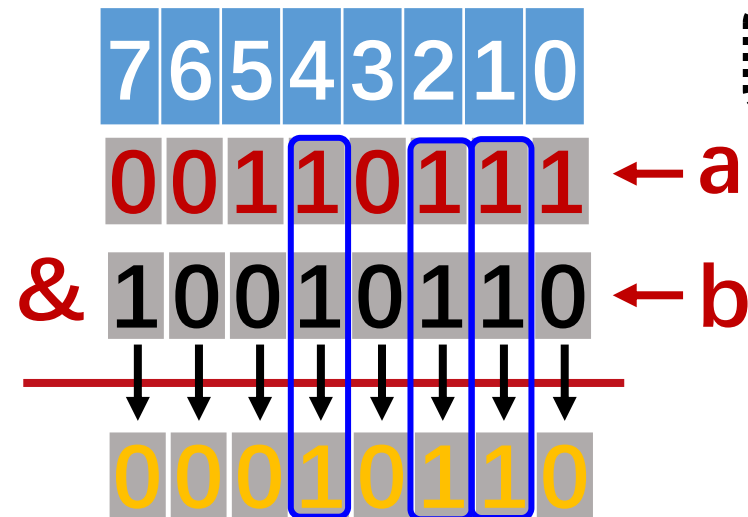
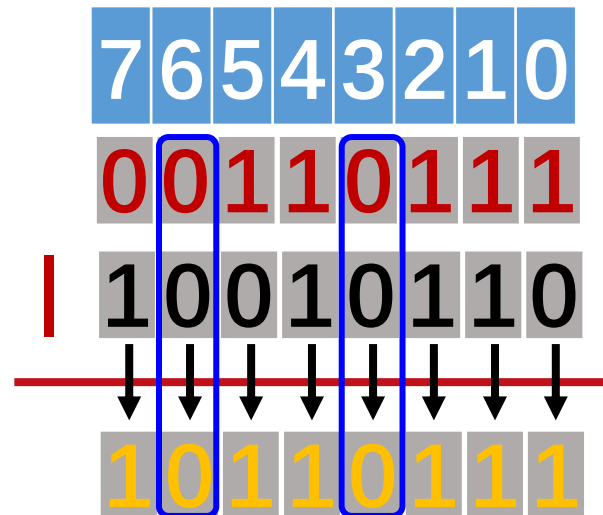
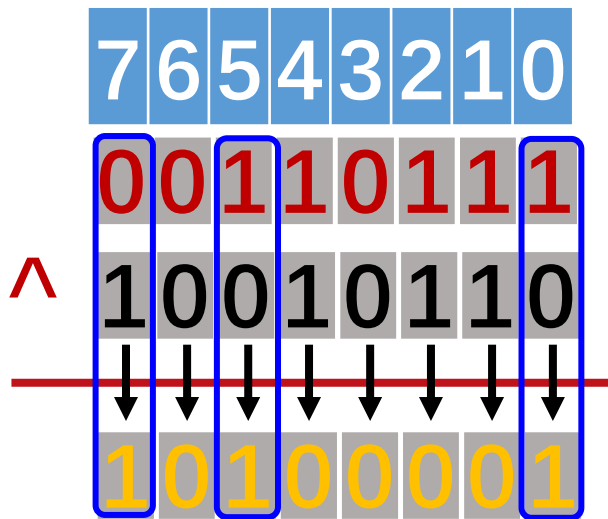
C语言的位运算

■二进制按位运算

C语言位运算

&, |, ^, ~

<<, >>



unsigned char a,b;

a=55

b=150

10进制

a=00110111

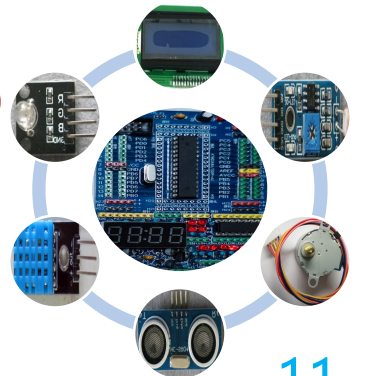
b=10010110

2进制

a=0x37

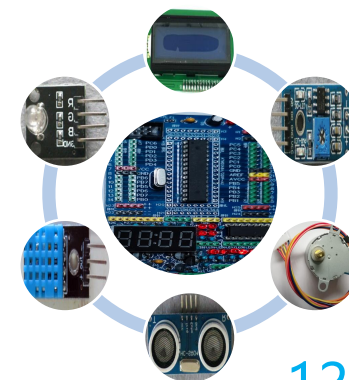
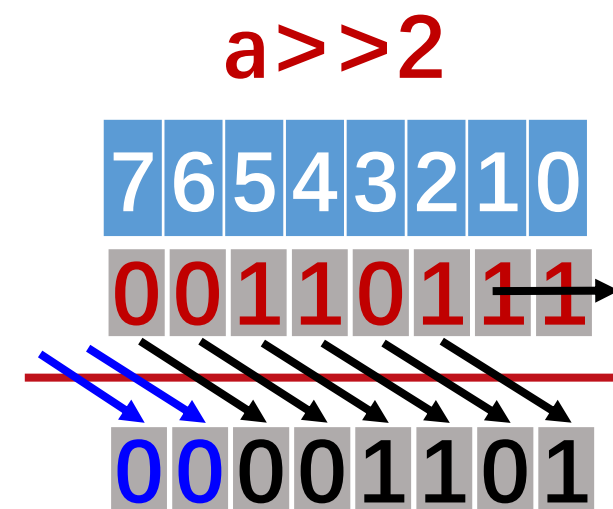
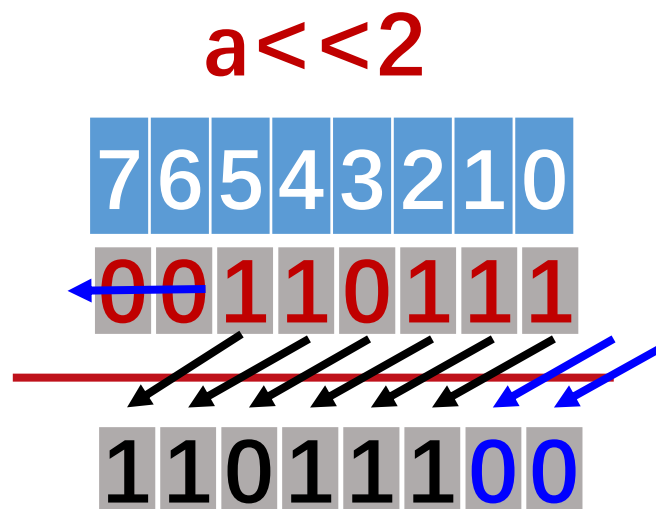
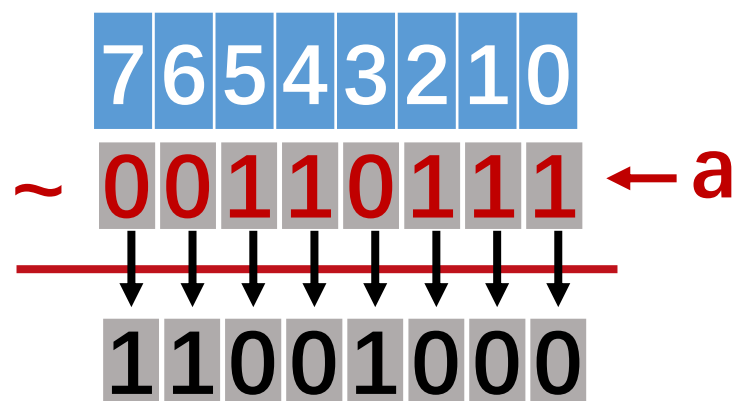
b=0x96

16进制



C语言的位运算

■二进制按位计算

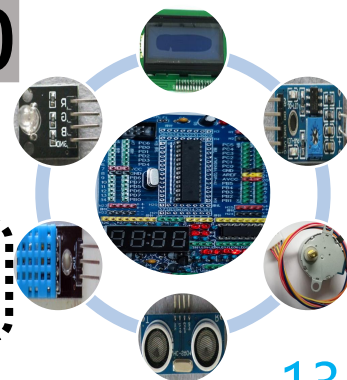
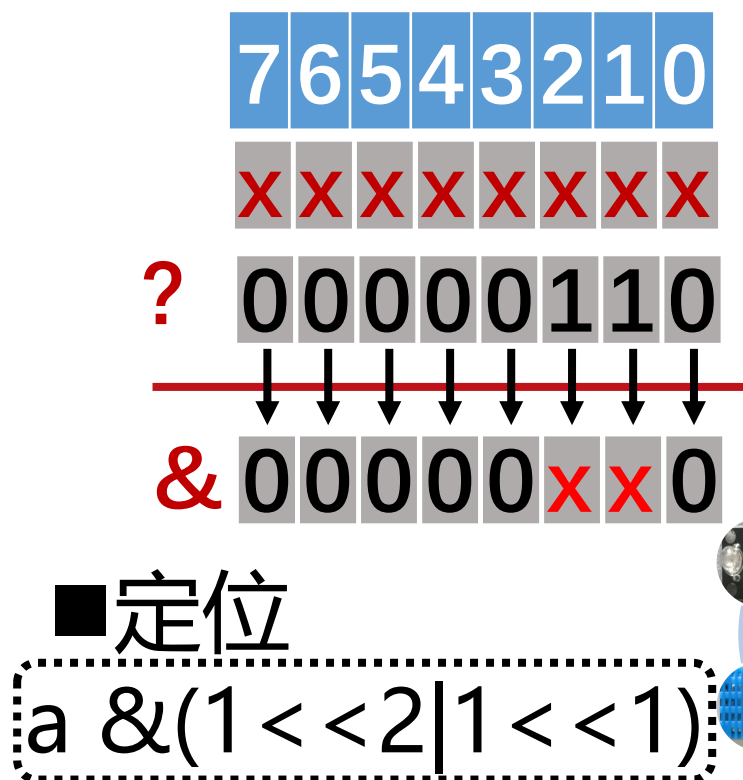
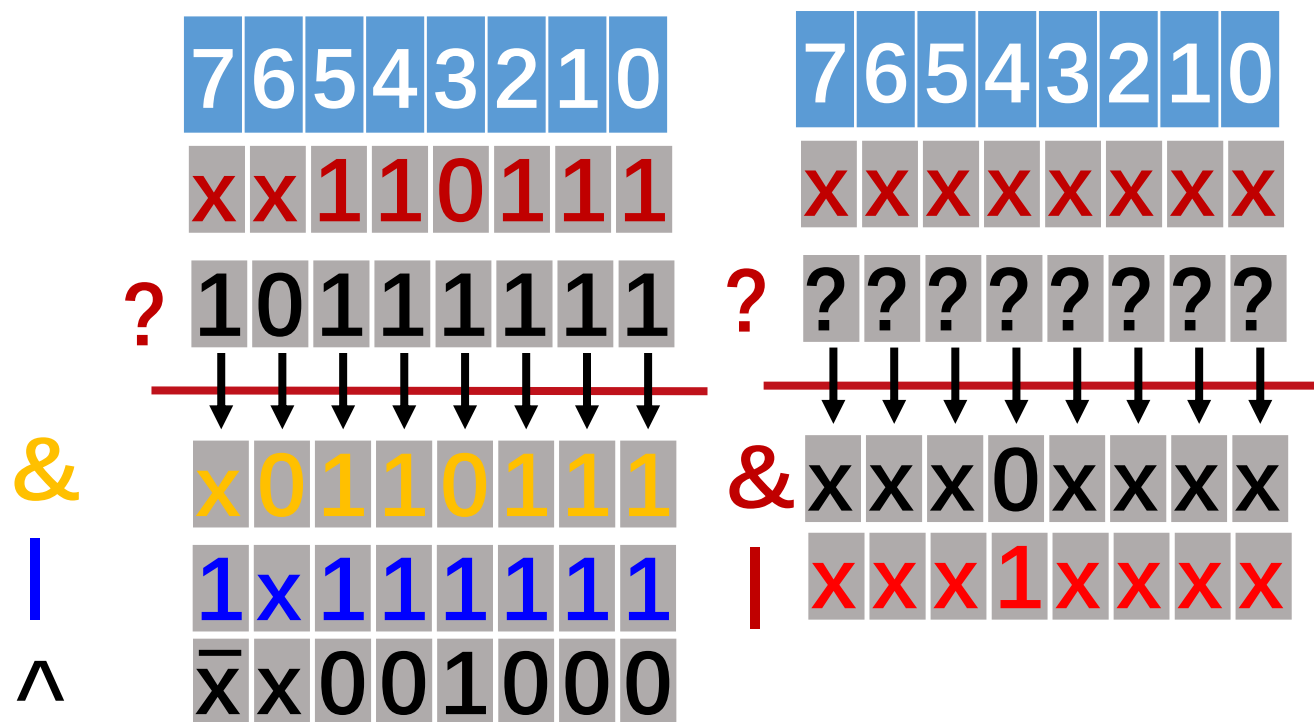


C语言位运算在单片机编程时的意义

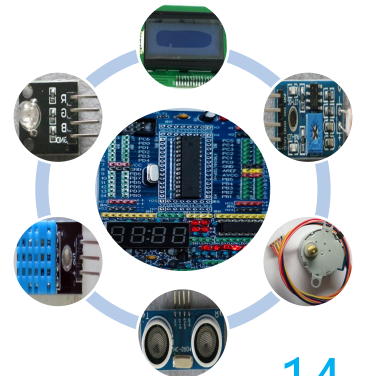
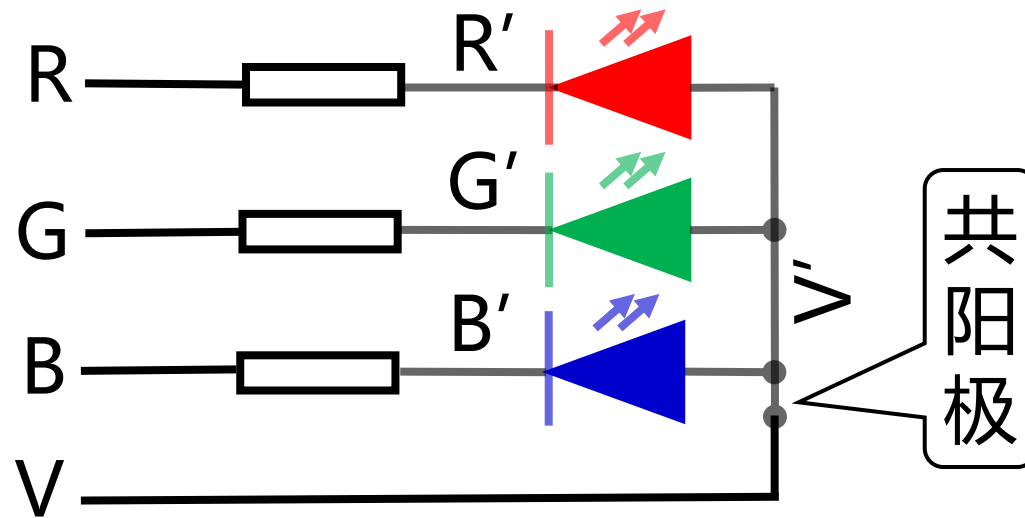
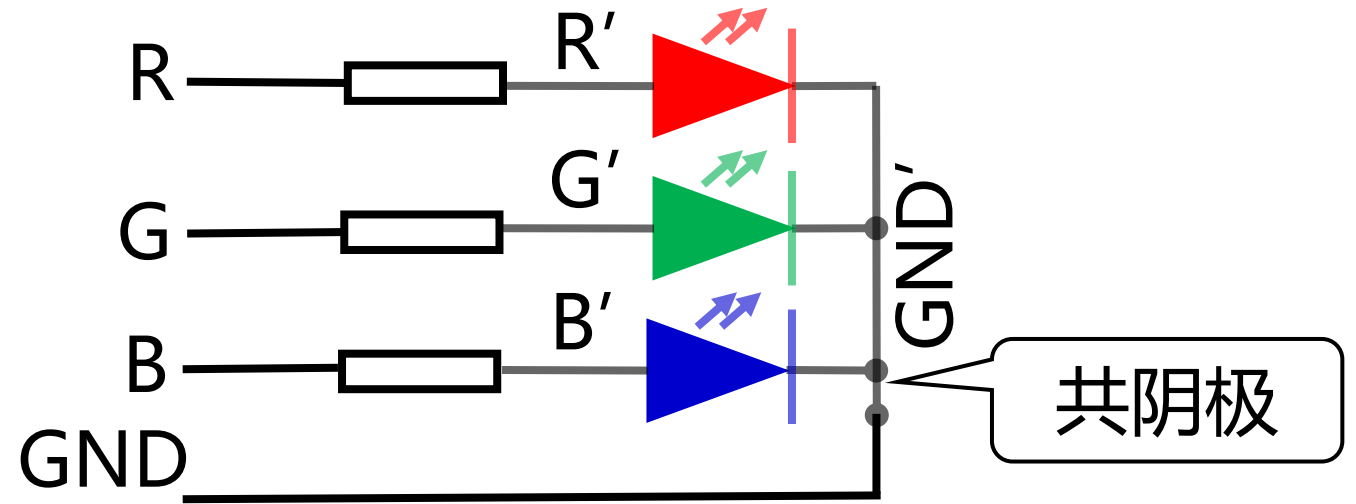
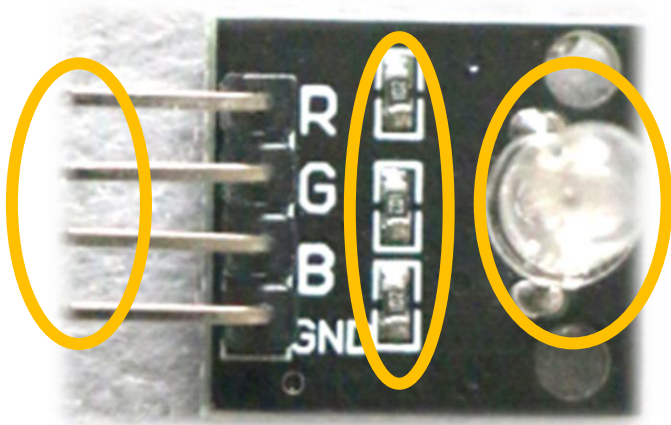
■改变指定位的值：清零/置位/取反

■改变指定位的值，保护其它位

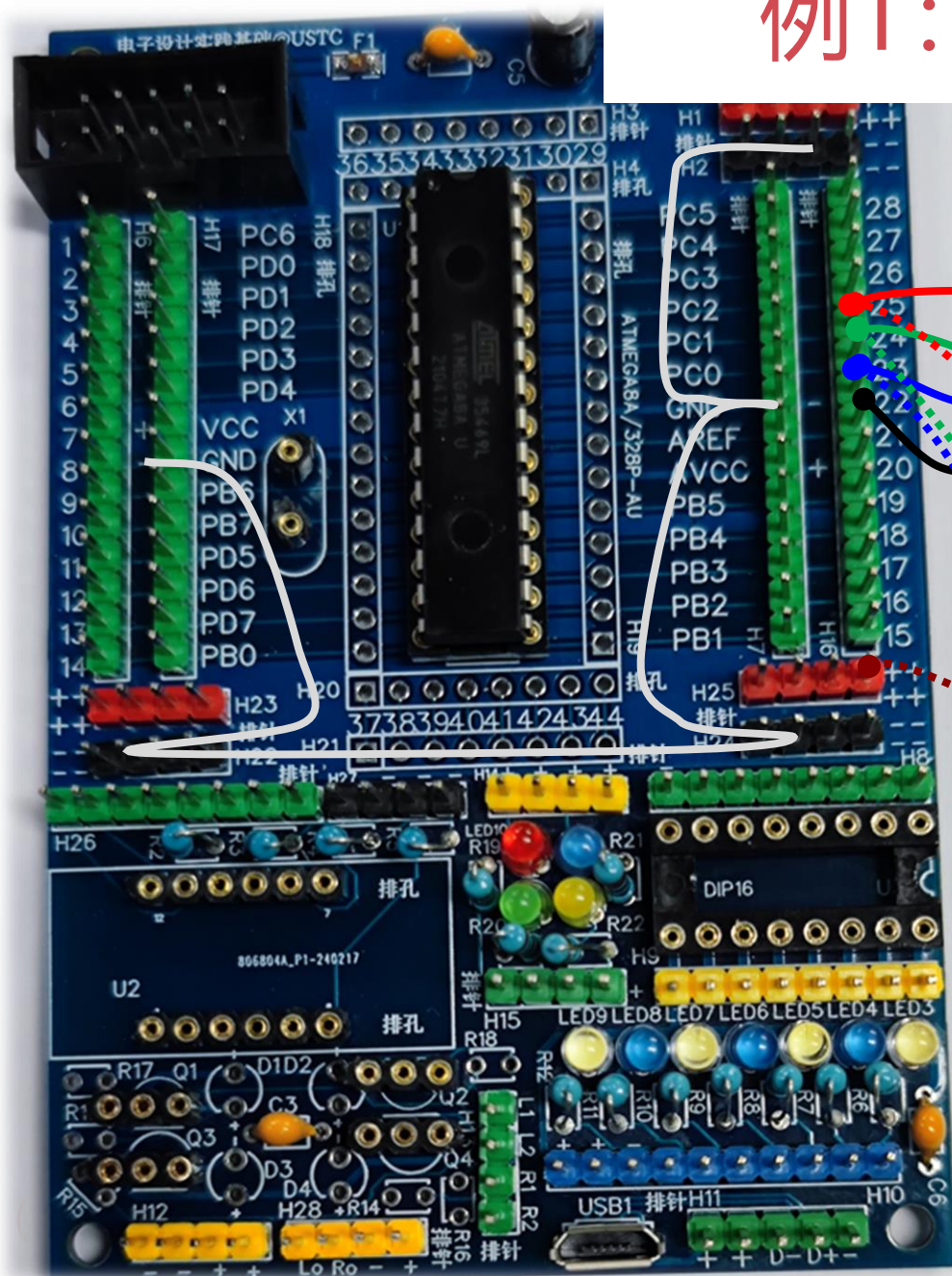
■取指定位的值



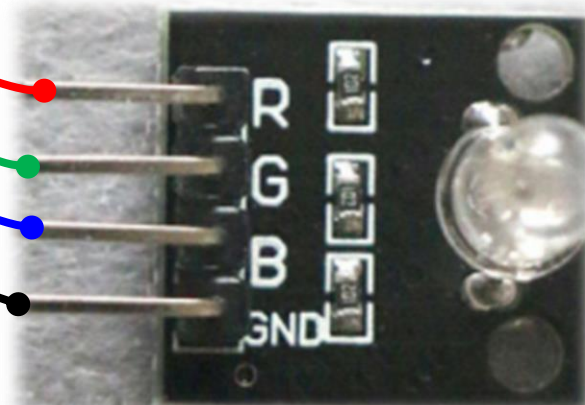
RGB LED模块



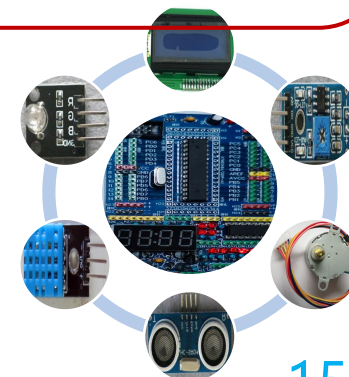
例1：控制RGB LED亮灭的硬件连接



由PC2/1/0管脚分别
控制LED的正极R/G/B



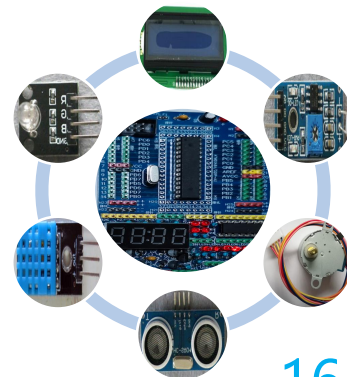
RGB LED与
MCU使用
同一电源的
正/负极



例1：控制RGB LED亮灭的编程关键

■流水灯：轮流点亮每一个颜色

```
int i,j;  
DDRC |= (1<<DDRC2) | (1<<DDRC1) | (1<<DDRC0);  
while (1)  
{ PORTC = (1<<PORTC2); //only red on  
  for(i=0;i<100;i++) for(j=0;j<1000;j++);  
  PORTC = (1<<PORTC1); //only green on  
  for(i=0;i<100;i++) for(j=0;j<1000;j++);  
  PORTC = (1<<PORTC0); //only blue on  
  for(i=0;i<100;i++) for(j=0;j<1000;j++);  
}
```



编程拓展：关于库函数的精确延时

```
int i,j;
DDRC = (1<<DDRC2)|(1<<DDRC1)|(1<<DDRC0);
while (1)
{
    PORTC = (1<<PORTC2);
    for(i=0;i<100;i++)
        for(j=0;j<1000;j++);
    PORTC = (1<<PORTC1);
    for(i=0;i<100;i++)
        for(j=0;j<1000;j++);
    PORTC = (1<<PORTC0);
    for(i=0;i<100;i++)
        for(j=0;j<1000;j++);
}
```

■使用库函数精确延时步骤

1

#define F_CPU 1000000UL

2

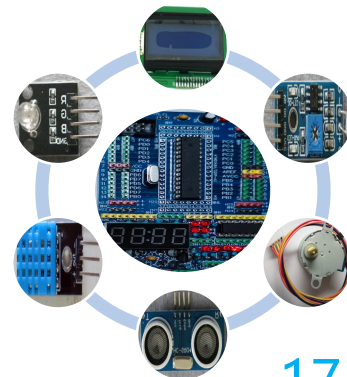
#include <util/delay.h>

3

_delay_ms(毫秒);

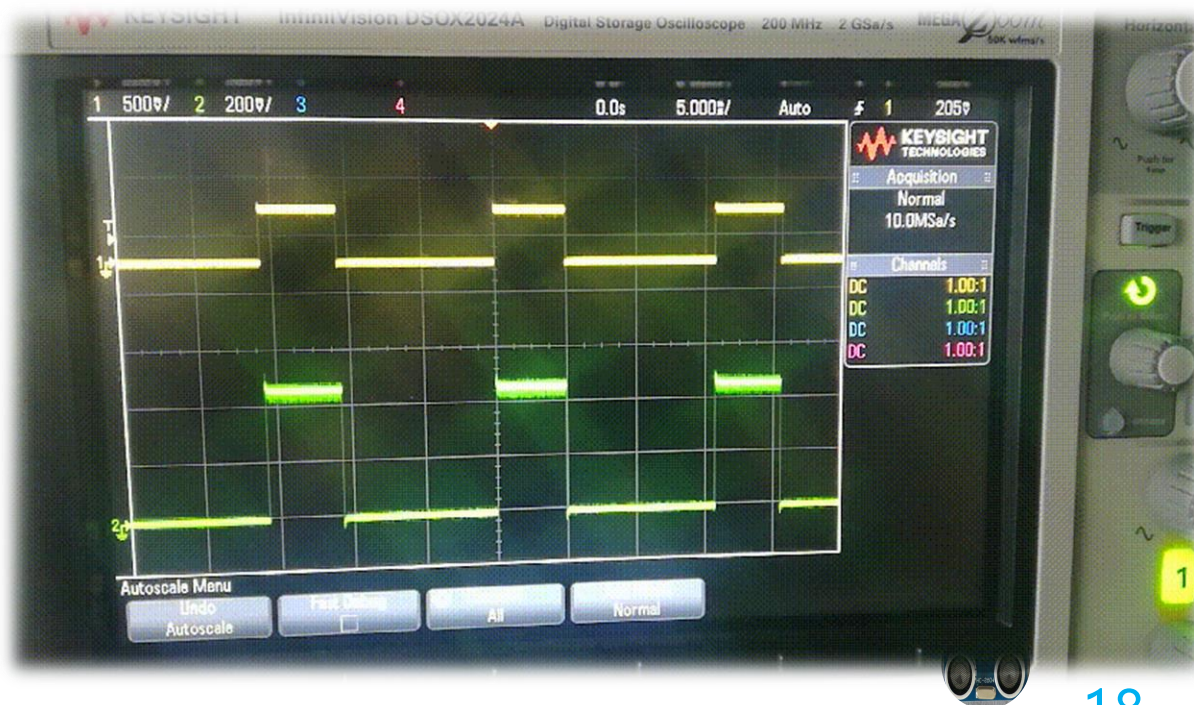
_delay_us(微秒);

常数



例2：LED呼吸灯的实现

- 灯的亮灭，模仿呼吸过程
- 整个过程约3秒，渐亮/渐灭各1.5秒
- 线性或非线性
- MCU实现原理：调整脉冲高/低电平的宽度，即PWM（脉冲宽度调制）
- LED余晖效应



例2：实现呼吸灯的编程关键

■在PC0管脚外接一个LED，以实现呼吸灯

```
int i,j;
```

```
DDRC = (1<<DDRC0); //PC0为输出
```

```
for(i=0;i<100;i++)//渐亮
```

```
{ PORTC |= (1<<PORTC0); //PC[0]=1
```

```
  for(j=0;j<=i;j++)_delay_us(150);
```

```
  PORTC &= ~(1<<PORTC0); //PC[0]=0
```

```
  for(j=0;j<100-i;j++)_delay_us(150);
```

```
}
```

$$150\mu s \times 100 \times 100 \approx 1.5s$$

```
for(i=100;i>0;i--)//渐暗
```

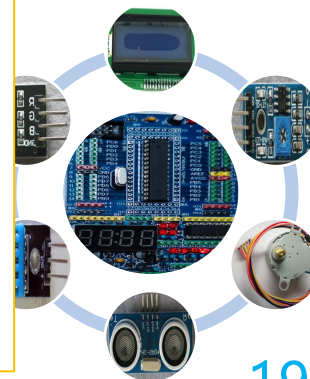
```
{ PORTC |= (1<<PORTC0); //PC[0]=1
```

```
  for(j=0;j<=i;j++)_delay_us(150);
```

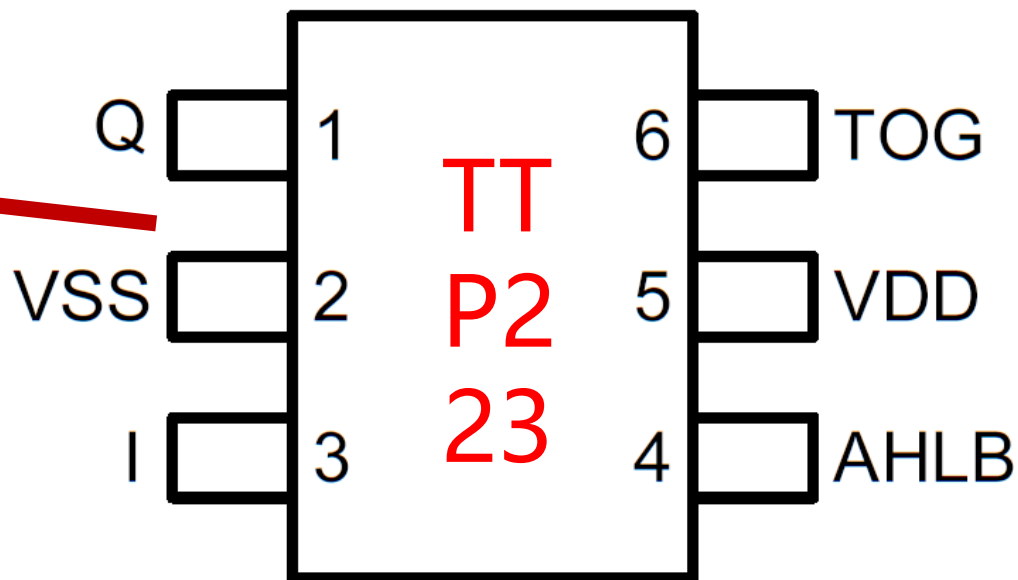
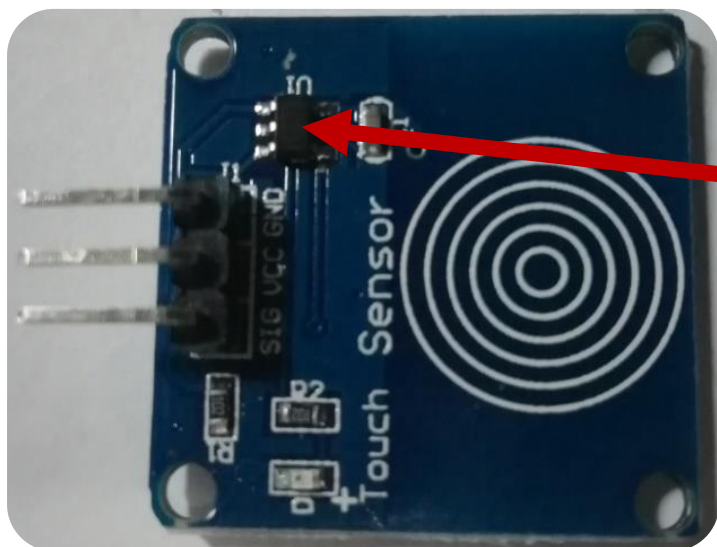
```
  PORTC &= ~(1<<PORTC0); //PC[0]=0
```

```
  for(j=0;j<100-i;j++)_delay_us(150);
```

```
}
```



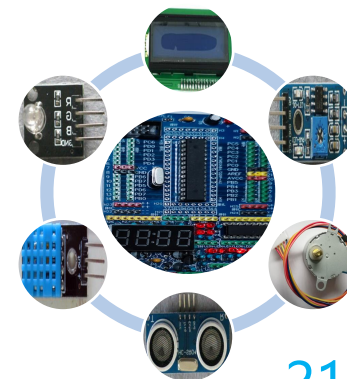
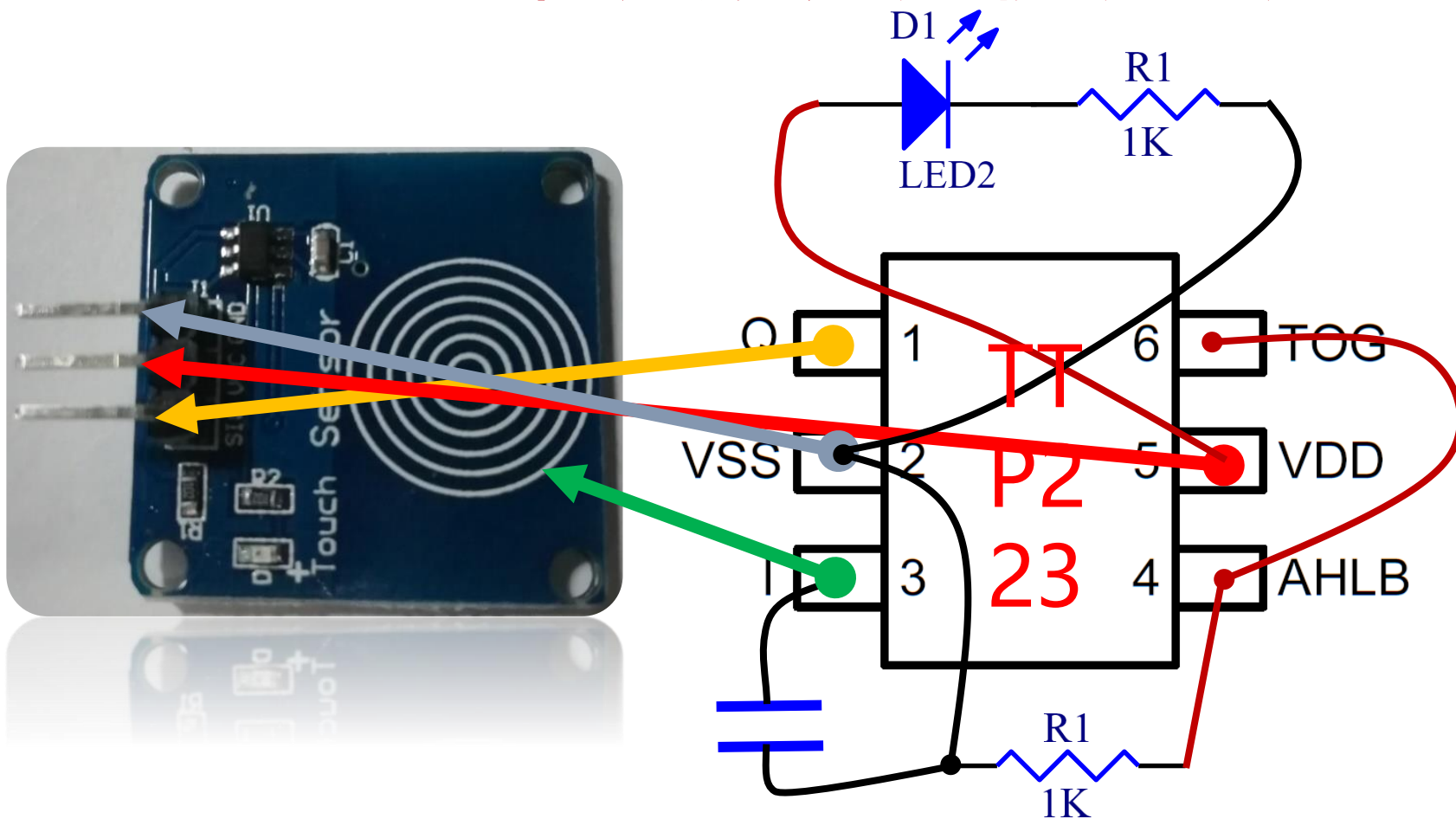
TTP223 (一位) 触摸开关模块



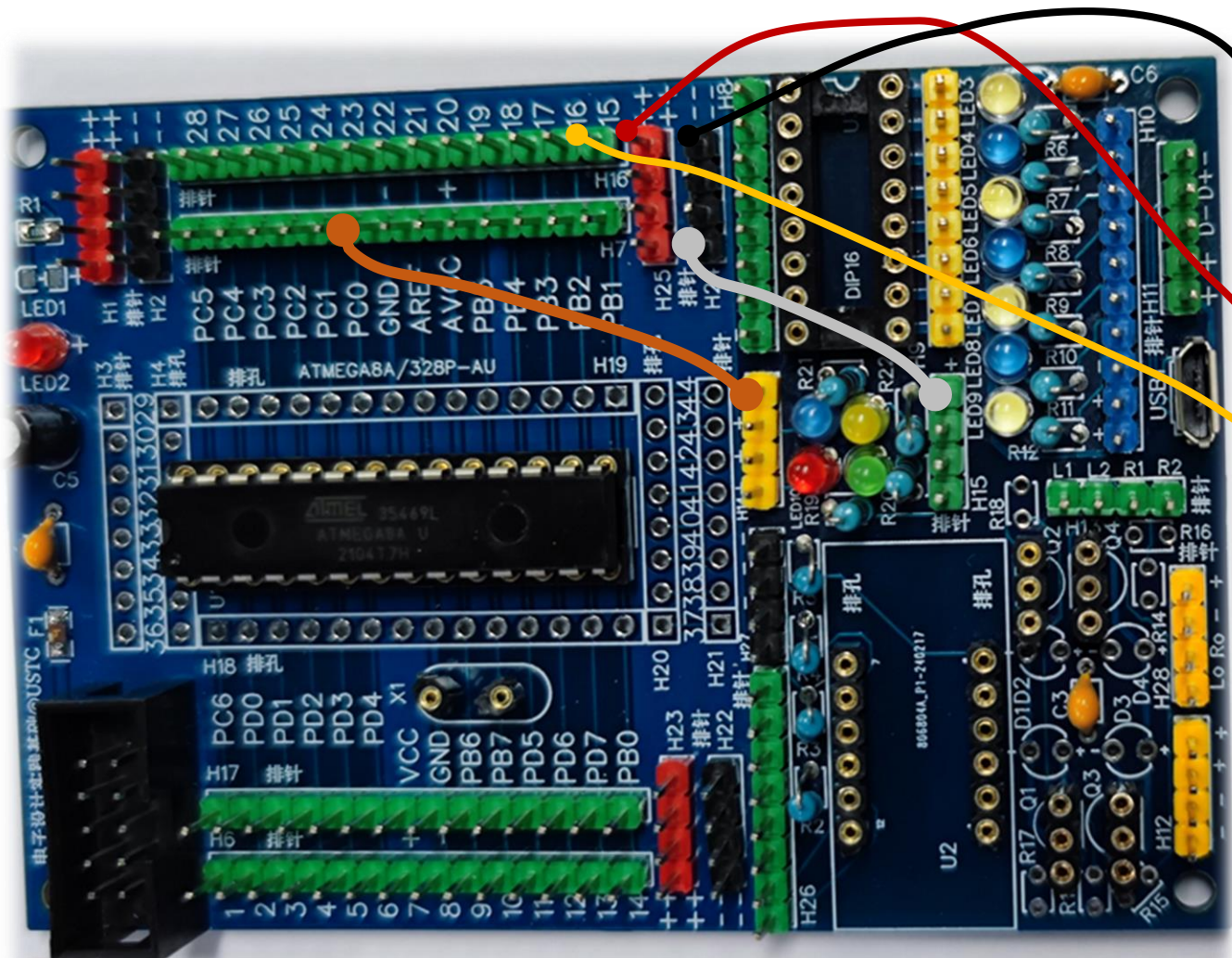
管脚编号	管脚名	IO类型	说明
1	Q	输出	CMOS输出管脚 (1/0)
5/2	VDD/VSS	电源	供电电源正/负极
3	I	输入/出	输入传感器端口
4	AHLB	输入	输出高/低有效电平选择:0-高,1-低
6	TOG	输入	输出类型选择: 0-直接, 1-切换

TTP223 一位触摸开关模块的原理

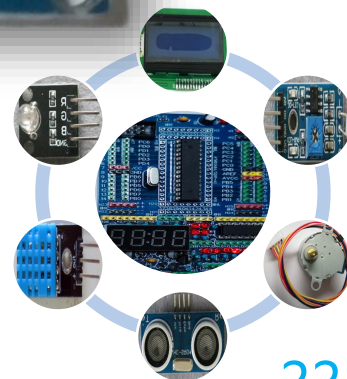
GND
VCC
SIG



例1：触摸开关控制LED亮灭的硬件连接

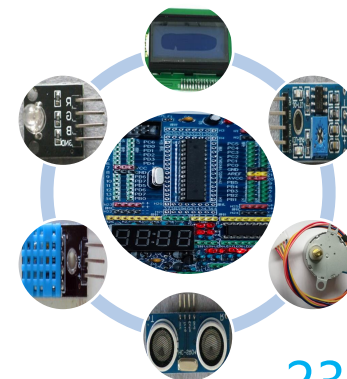


与MCU使用同一电源

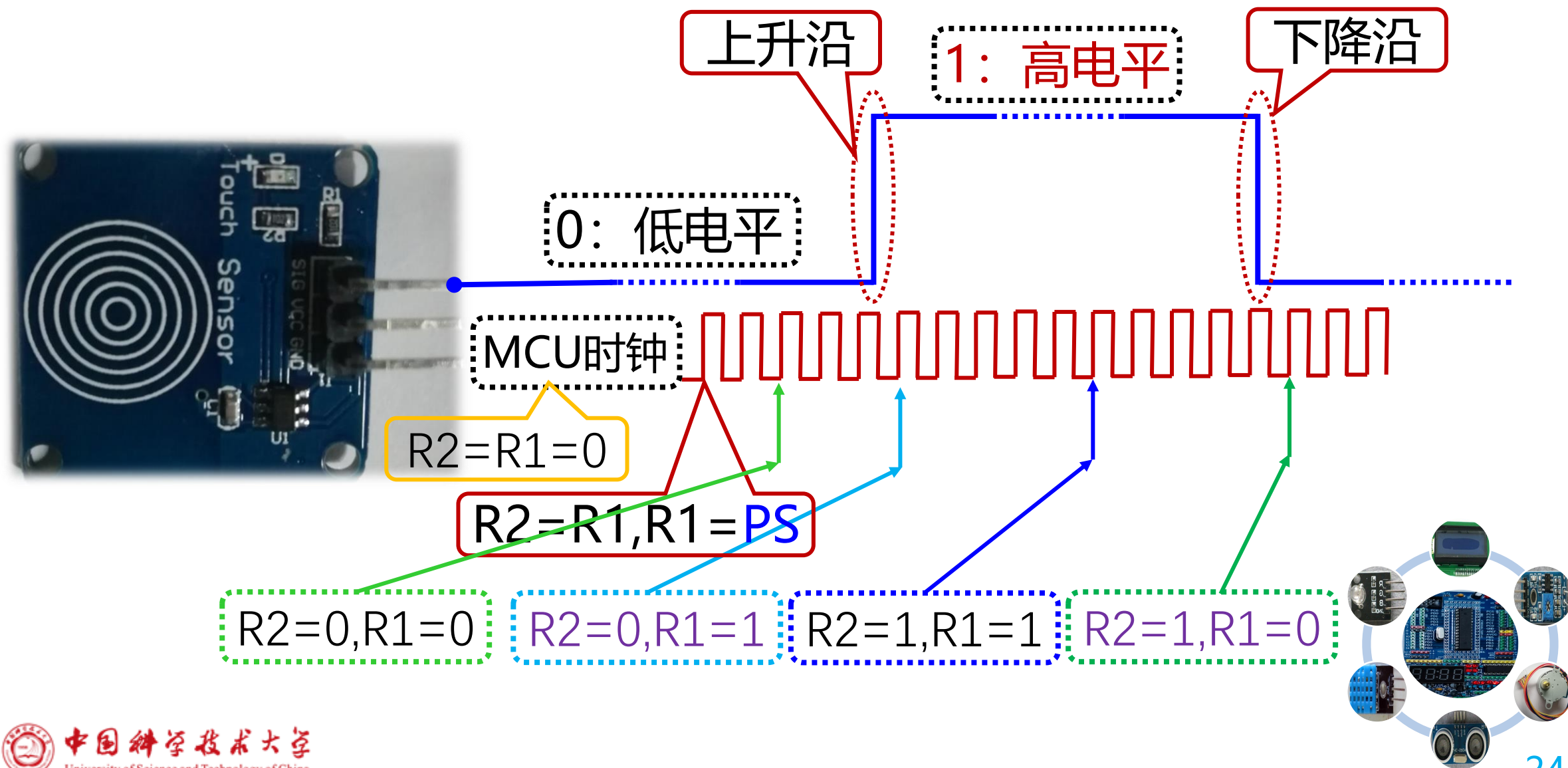


例1：触摸开关控制LED亮灭的编程关键

```
DDRC |= (1<<DDRC0); //PC0为输出
DDRB &= ~(1<<DDRB1); //PB1为输入
while (1)
{
    if(PINB & (1<<PINB1)) //PINB[1]=1?
        PORTC |= (1<<PORTC0); //PORTC[0]=1
    else
        PORTC &= ~(1<<PORTC0); //PORTC[0]=0
}
```



知识拓展：如何统计触摸开关的开关次数



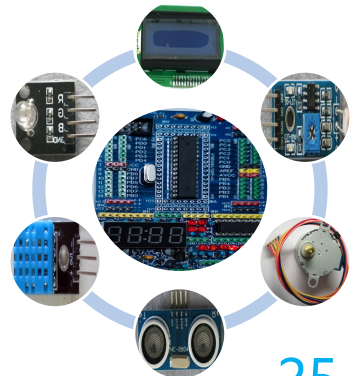
例2：用TTP223控制RGB LED以不同方式亮灭

■用接在PB1管脚的触摸开关控制PC0/1/2管脚上的RGB LED的亮灭

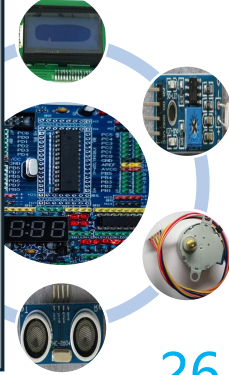
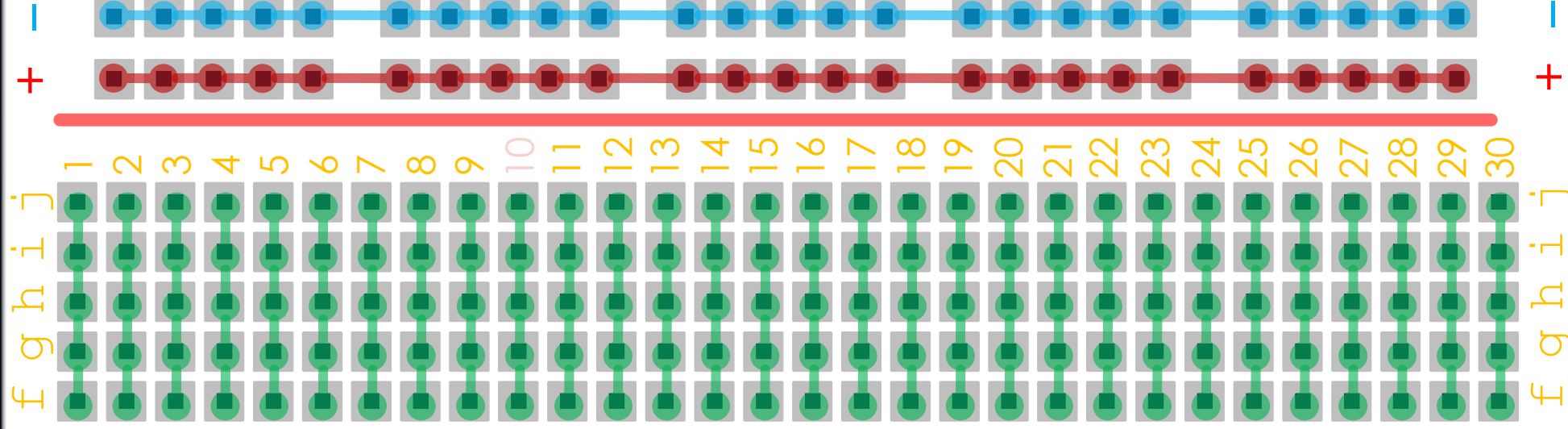
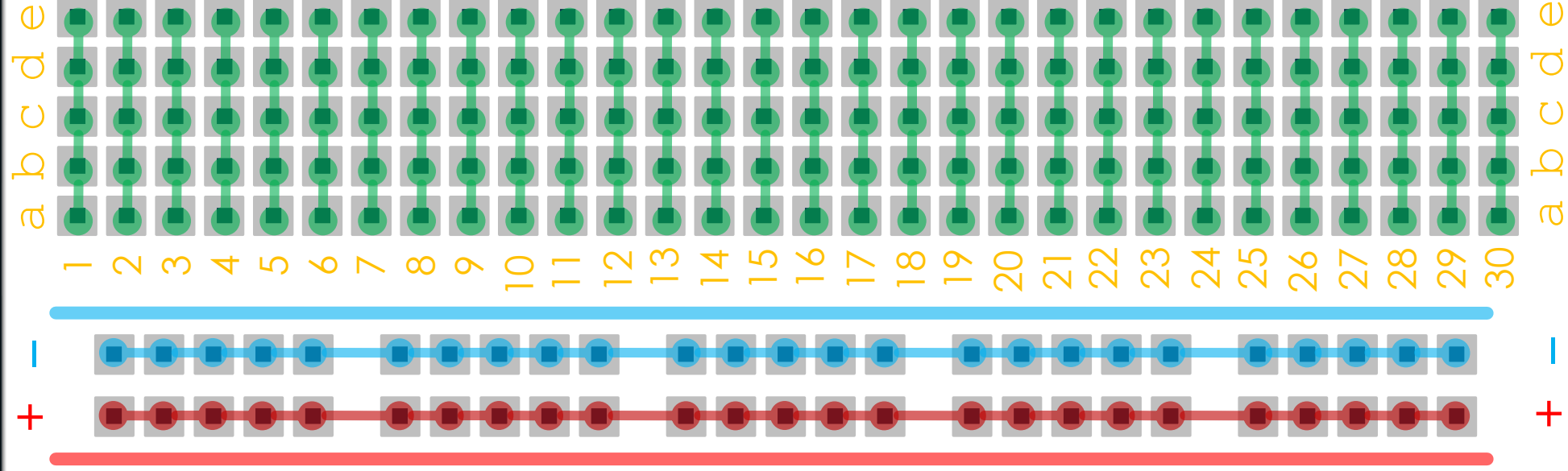
```
unsigned char tpr1=0,tpr2=0,counter=0;  
DDRC = (1<<DDRC2)|(1<<DDRC1)|(1<<DDRC0); //PC0/1/2为输出模式  
DDRB &= ~(1<<DDRB1); //PB1为输入
```

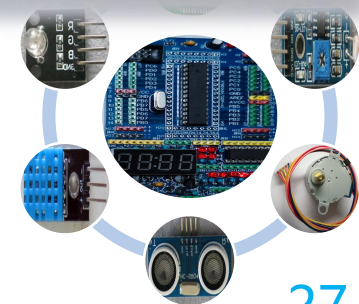
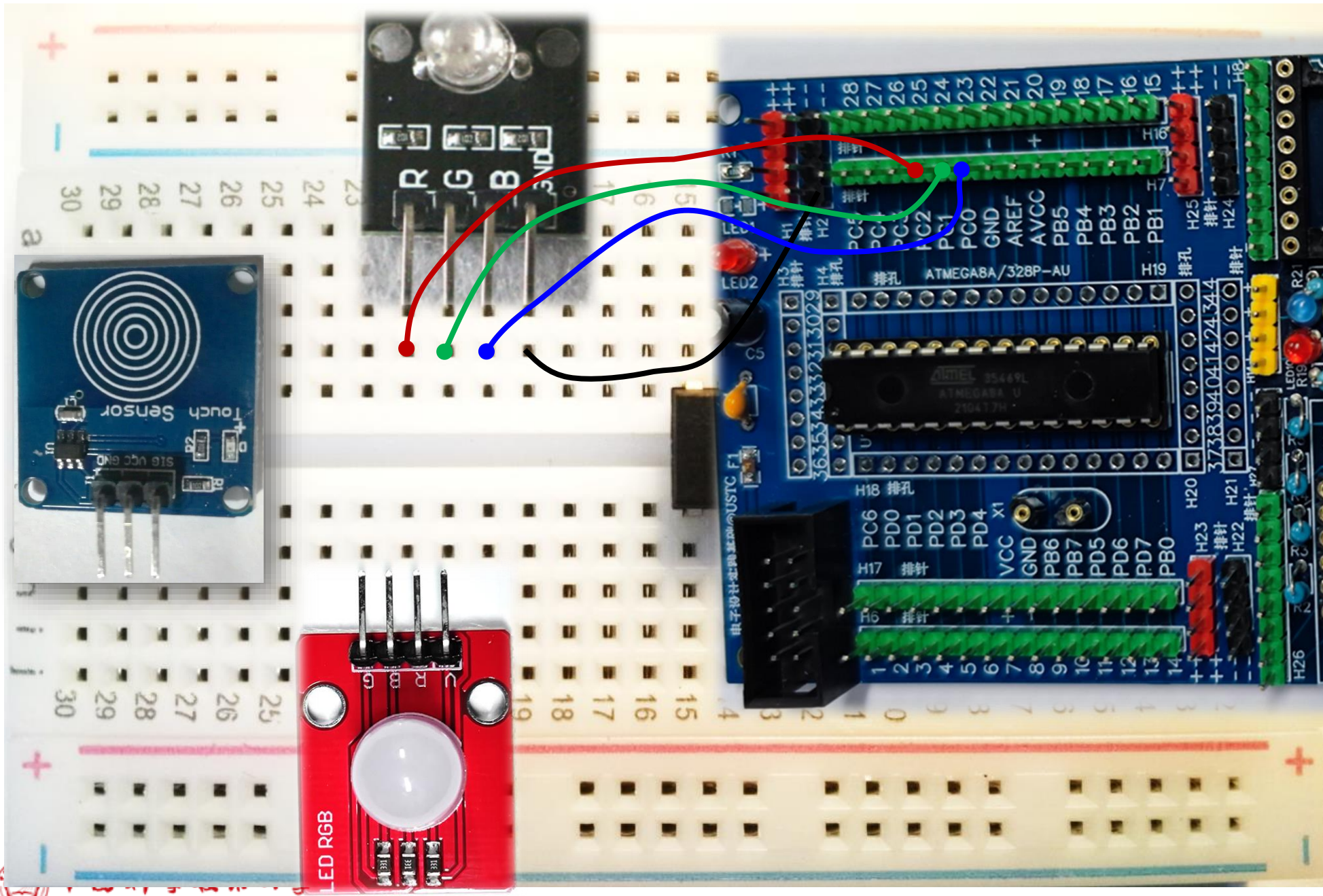
while(1)

```
tpr2 = tpr1;  
tpr1 = (PINB & (1<<PINB1))>>1;  
if(tpr2==0 && tpr1==1)counter++;  
switch(counter)  
{  
case 1 : PORTC = (1<<PORTC2);break;  
case 2 : PORTC = (1<<PORTC1);break;  
case 3 : PORTC = (1<<PORTC0);break;  
default :PORTC =0;counter = 0;  
}
```



面包板





本周实验内容

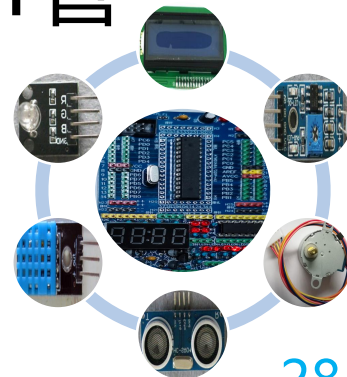
实验内容1：用**PB3/2/1**分别控制RGB LED中的R,G和B正极，实现**5种颜色以上**的循环亮灭

实验内容2：利用**PB2**管脚实现**蓝色**的**呼吸灯**

实验内容3：用接在**PD4**管脚外的**触摸开关**去控制**PB2**管脚外连接的LED按**呼吸灯**方式亮灭

实验内容4：用接在PD4管脚的触摸开关控制PB3,2,1管脚上的RGB LED的亮灭

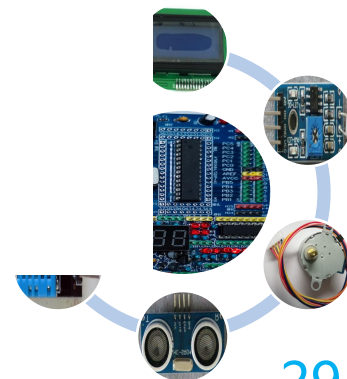
当次全部完成后当场验收，总结下次交



实验注意事项与应急处理



- 连接下载线到电脑USB接口时应注意：
 - 实验板**电源指示灯**是否亮，如**不亮**可能存在连接短路，须**立即断开**下载线与电脑USB接口的连接
- 打开下载程序PROGISP (Ver 2.0)
 - 确认下载线连接正确（连接状态图标为彩色）
 - 确认**芯片型号**正确，调入Flash程序正确等



谢谢

